

| parameter                   | UCI                                         | BU   | notes               |
|-----------------------------|---------------------------------------------|------|---------------------|
| N.B. when t                 |                                             |      |                     |
| <b>p.screen</b>             |                                             |      | <b>screen para</b>  |
| distFromScreen_inCm:        | 75                                          | 75   |                     |
| screenWidth_inPixels:       | 1280                                        | 1920 |                     |
| screenHeight_inPixels:      | 1024                                        | 1080 |                     |
| screenWidth_inCm:           | 40                                          | 50   |                     |
| screenHeight_inCm:          | 30                                          | 30   |                     |
| refreshRate_inHz:           | 60                                          | 120  |                     |
| contrastResolution          | 0.0078                                      |      | minimum p           |
| <b>p.stim</b>               |                                             |      | <b>general stir</b> |
| apertureWidth_inDeg:        | 4                                           |      | "aperture" i        |
| fixationWidth_inDeg:        | 0.35                                        |      | fixation cro        |
| cueWidth_inDeg:             | 0.1                                         |      | cue with in         |
| cueLength_inDeg:            | 1                                           |      | cue length i        |
| cueBuffer_inDeg:            | 0.5                                         |      | distance b/         |
| BGcolor:                    | 127                                         |      | background          |
| cueColorInactive:           | 0                                           |      | color of ina        |
| cueColorActive:             | 255                                         |      | color of act        |
| <b>p.stim.periph</b>        |                                             |      | <b>peripheral</b>   |
| stimDistFromCenter_inDeg:   | 5                                           |      | distance frc        |
| ovalMajorAxis_inDeg         | 5                                           |      | length of th        |
| ovalMinorAxis_inDeg         | 5                                           |      | length of th        |
| circleWdith_inDeg           | 5                                           |      | width of a c        |
| <b>p.stim.gab</b>           |                                             |      | <b>peripheral</b>   |
| gab.aperture                | cosine'                                     |      | grating ape         |
| gab.apertureEdgeWidth_inDeg | 0.5                                         |      | grating ape         |
| gab.apertureEdgeWidth_inPix | 25                                          |      | grating ape         |
| gab.rad                     | 100                                         |      | grating radi        |
| gab.noiseType               | filteredSF'                                 |      | type of noi         |
| gab.noiseContrast           | 0.5                                         |      | noise contr         |
| gab.gratingSF               | 1                                           |      | grating spat        |
| gab.orientation             | [-45, 45]                                   |      | list of possi       |
| gab.phase                   | 0                                           |      | grating pha         |
| gab.gratingContrast_min     | 0.05                                        |      | minimum p           |
| gab.gratingContrast_max     | 0.2                                         |      | maximum p           |
| gab.contrastPrior_list      | [0.044 0.082 0.098 0.112 0.127 0.146 0.270] |      | list of 7 pos       |
| gab.contrast_list           | individually thresholded                    |      | list of 7 pos       |
| <b>p.timing</b>             |                                             |      | <b>duration of</b>  |
| fixation1Dur_inSecs:        | 0.5                                         |      | initial fixati      |
| fixation3Dur_inSecs:        | 0.15                                        |      | next fixatio        |
| precueDur_inSecs:           | 0.05                                        |      | activation c        |
| prePeriphStimDur_inSecs:    | 0.25                                        |      | next fixatio        |

|                           |                                          |                |
|---------------------------|------------------------------------------|----------------|
| periphStimDur_inSecs:     | 0.25                                     | presentation   |
| fixation4Dur_inSecs:      | 0.25                                     | next fixation  |
| postcueDur_inSecs:        | 5                                        | activation c   |
| ITIDur_inSecs:            | 0.5                                      | time betwe     |
| <b>p.eyetracker.trial</b> |                                          |                |
| fixationRadius_inDeg:     | 2                                        | fixation acc   |
| fixationDur_inSecs:       | 0.5                                      | radius of th   |
| timeoutDur_inSecs:        | 10                                       | length of ti   |
|                           |                                          | if fixation is |
| <b>p.eyetracker.block</b> |                                          |                |
| fixationRadius_inDeg:     | 1                                        | fixation acc   |
| fixationDur_inSecs:       | 1                                        | radius of th   |
| timeoutDur_inSecs:        | 10                                       | length of ti   |
|                           |                                          | if fixation is |
| <b>p.kb</b>               |                                          |                |
| exitKey:                  | 'ESCAPE'                                 | keyboard e     |
| respKeys:                 | {'1!' '2@' '3#' '8*' '9(' '0)' 'ESCAPE'} | key used to    |
|                           |                                          | valid keys fo  |
|                           |                                          | 1 - saw grati  |
|                           |                                          | 2 - saw grati  |
|                           |                                          | 3 - saw grati  |
|                           |                                          | 8 - saw grati  |
|                           |                                          | 9 - saw grati  |
|                           |                                          | 0 - saw grati  |
| <b>p.auditoryFB</b>       |                                          |                |
| toneFreq:                 | [523, 784]                               | auditory fe    |
| toneDur_inSecs:           | 0.25                                     | tone freque    |
|                           |                                          | tone durati    |
| <b>p.quest</b>            |                                          |                |
| pThreshold:               | 0.75                                     | QUEST thre     |
| tGuess:                   | -0.9208                                  | target value   |
| tGuessSD:                 | 3                                        | initial gues   |
| beta:                     | 3.649                                    | SD of tGues    |
| delta:                    | 0.039                                    | slope of the   |
| gamma:                    | 0.5                                      | fraction of t  |
| grain:                    | 0.01                                     | chance rate    |
| range:                    | 5                                        | quantizatio    |
|                           |                                          | the intensit   |

the BU column is left empty, the value used is the same as the one listed in the UCI column

## Parameters

possible contrast

### Stimulus parameters

refers to the gray circle in which the fixation cross and cues are displayed

circle width

in degrees

in degrees

distance from fixation center and cue edge

background color when no line texture stim is shown

inactive cues. Cues are inactive when drawn on the screen but not actively directing attention or response

active cues. Cues are active when directing attention (precue) or response (postcue)

### Stimulus parameters

distance from center of screen to center of peripheral stimulus

angle of peripheral stimulus along its major axis (in degrees)

angle of peripheral stimulus along its minor axis (in degrees)

radius of circle whose area equals the area of the peripheral stimulus with major and minor axes as specified above (in degrees)

### Grating and noise parameters

texture type

texture edge width (in degrees)

texture edge width (in pixels)

radius (in pixels), equal to  $\text{periph.circleWidth\_inPix}/2 - \text{gab.apertureEdgeWidth\_inPix}$

radius, filtered around grating spatial frequency

radius

spatial frequency (in cycles per degree)

number of grating orientations (in degrees)

radius

possible grating contrast (for determining priors only)

possible grating contrast (for determining priors only)

possible grating contrasts. This is a pre-defined prior based on pilot data.

possible grating contrasts. Before thresholding, this is a pre-defined prior based on pilot data. After thresholding

### Each trial stage

duration. If eye tracking, this is the \*minimum\* duration of this trial stage (but may be longer depending on time)

in stage (inactive cues appear)

duration of the precue (active cue changes color)

in stage (all cues return to inactive status)

n of the peripheral noisy grating stimuli  
n stage (all peripheral stimuli removed, only fixation and inactive cues remain)  
of the postcue (active cue changes color). This is the \*maximum\* duration for this trial stage; in practice, this  
en keypress on trial N and onset of initial fixation for trial N+1

#### **Fixation prior to each trial**

the circle in which fixation must remain during fixation acquisition  
the time in which fixation must remain in the fixation region during fixation acquisition  
if not acquired within this time limit, the subject is shown a text prompt and fixation acquisition resets. After

#### **Fixation prior to each block (more stringent than pre-trial fixation)**

the circle in which fixation must remain during fixation acquisition  
the time in which fixation must remain in the fixation region during fixation acquisition  
if not acquired within this time limit, the subject is shown a text prompt and fixation acquisition resets. After

#### **Entry of responses**

to exit the experiment  
for response entry during the experiment. Meanings of the keys are as follows:  
1: saw orientation, saw oriented -45 from vertical  
2: did not see orientation, guess oriented -45 from vertical  
3: did not see orientation, guess oriented -45 from vertical  
4: did not see orientation, guess oriented +45 from vertical  
5: did not see orientation, guess oriented +45 from vertical  
6: saw orientation, saw oriented +45 from vertical

#### **Feedback during practice**

frequency (in Hz) following incorrect and correct responses, respectively  
duration in seconds

#### **Thresholding**

percentage of p(correct) for thresholding in the oval orientation discrimination task  
percentage of the line length threshold, based on previous pilot data collection  
percentage, using default / recommended value  
percentage of psychometric function, based on previous pilot data collection  
percentage of trials on which the subject presses blindly (determines upper asymptote of the psychometric function). Based on  
percentage of p(correct)  
percentage (step size) of the internal table, using default / recommended value  
percentage of the difference between the largest and smallest intensity that the internal table can store, using default / recommended



degrees)

ng, this list is updated based on QUEST results.

needed for fixation acquisition)

s trial stage ends as soon as the subject enters their response. If no response is entered within the maximum

er 2 consecutive timeouts, the subject is prompted to get the experimenter to help with eyetracker calibrat

er 2 consecutive timeouts, the subject is prompted to get the experimenter to help with eyetracker calibrat

ed on previous pilot data collection

mmended value



time limit, the trial stage ends automatically

ion

ion